

ITSS SOFTWARE DEVELOPMENT
SOFTWARE DESIGN AND CONSTRUCTION
0. INTRODUCTION TO COURSE

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Class rules

- **Before Class**

- ✓ Be fully prepared for the lesson
- ✓ Ensure the board is clean
- ✓ Bring your notebook/computer
- ✓ Arrive on time
- ✓ No need to ask for permission to enter or leave

- **During Class**

- 🤫 Maintain silence
- 🚫 Avoid using computers or mobile phones during lectures
- 📝 Participate in exercises and quizzes

- **After Class**

- 📖 Work on case studies
- 🎓 Complete the capstone project

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Course objectives

- **Design** effective program structures with
 - appropriate modularity
 - separation of abstraction and implementation concerns
 - use of standard design patterns
- Use modern **programming** languages and **unit test** framework effectively
 - type systems, objects and classes, modularity
 - identity and equality, exceptions and assertions
- Gain experience with contemporary software tools
 - design tools
 - integrated development environments (IDE)
 - test frameworks, version control
 - documentation processing tools

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Main topics

- Software Development Process
- Object-Oriented Analysis and Design
- Design Concepts: Coupling and Cohesion
- Design Principles, Design Patterns
- Software Quality, Software Testing
- Techniques for Effective Programming

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Programming language/tools

- Software design tool: Astah
 - Free for students
- Programming language: Java
- IDE: Eclipse
- Version control: Github
- Test framework: JUnit
- Architectural model / pattern: 3 tiers / MVC



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Assessment

- Mid-term score: 50%
 - Exercise/Quiz: 40%
 - Case Study/Capstone Project: 60%
 - Work in groups, but individual score
 - *Reward or Punish*
 - *Participant at class: ± 1*
- Final score: 50%
 - Multiple-choice
 - Written exam

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Reference books

- D. Budgen. **Software Design**, 2nd Edition. Addison-Wesley. 2004.
- Cay Horstmann. **Object-Oriented Design and Patterns**. John Wiley & Sons, Inc. 2006
- Joshua Bloch. *Effective Java*, 2nd ed. Addison-Wesley, 2008
- Boris Beizer. **Software Testing** Techniques, 2nd Edition. International Thomson Computer Press
- Erich Gamma, Richard Helm, Ralph Johnson, John Vlissides. **Design Patterns: Elements of Reusable Object-Oriented Software**. 1994. Addison-Wesley Professional.
- Robert Martin. **Agile Software Development: Principles, Patterns, and Practices**. 2003. Pearson Education.
- Robert Martin. **Clean Code: A Handbook for Agile Software Craftsmanship**. 2008. Prentice Hall.

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Course Materials

- Lecture notes for students (pdf): Slides in 4-page handouts
- Assignments, Project descriptions
- Interaction channels:
 - MS Teams
 - <https://github.com>
 - Add to your project member:
trangntt.for.student@gmail.com

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Naming convention

- <https://github.com>
 - Add to your project member:
trangntt.for.student@gmail.com
- Naming your private group repo (Capstone Project)
 - **TeamName-GroupNo**
 - E.g
 - [TKXDPM.20251-12](#)
 - [TKXDPM.VP.20251-05](#)
 - Add all members to the repo
- Monitor:

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Introduce yourselves

- Full name
- Experience in Computer Science
 - Operating System
 - Programming Languages
 - (Mini-)Projects
 - ...
- Strength / Weakness
- A course you like best / hate
- Desire to study in this course



About Me

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